

PREDICTIVE MODELLING OF HUMAN BODY FAT PERCENTAGE USING REGRESSION ANALYSIS

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Machine Learning – Course Project Report

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1. PROBLEM STATEMENT

Body fat percentage (BF %) is a key physiological indicator that reflects health status, body composition, and the risk of obesity-related diseases. While methods like DEXA scanning and hydrostatic weighing are precise, they are costly, inconvenient, and unsuitable for large-scale applications. Hence, estimating BF % through simple, easily measurable variables such as body weight, height, and circumferences offers a more practical alternative.

The goal of this project is to construct a regression-based predictive model capable of accurately estimating an individual's **body fat percentage** using multiple anthropometric measurements. Two regression techniques are used and compared:

1. **Linear Regression (Ordinary Least Squares)** – assumes a direct linear relationship between predictors and target.
 2. **Ridge Regression (L2 Regularization)** – introduces a penalty term that constrains large coefficients, thus reducing overfitting and improving stability when predictors are correlated.
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2. METHODOLOGY

2.1 Dataset Source and Preprocessing

The dataset used in this project is the *Body Fat Prediction Dataset* (Fede Soriano, Kaggle, 2023). It consists of **252 samples and 14 variables**, where **Body Fat (%)** is the dependent variable. Each record corresponds to one individual's anthropometric profile.

All rows containing missing or invalid entries were removed. The remaining features were numeric, making them suitable for regression.

2.2 Data Splitting and Scaling

The data was split as follows:

- **Training Set:** 70%
- **Validation Set:** 10%
- **Testing Set:** 20%

To ensure balanced feature contribution, all predictors were **standardized** using z-score normalization:

$$z = \frac{x - \mu}{\sigma}$$

This transformation prevents features with large scales (e.g., abdomen circumference in cm) from dominating smaller ones (e.g., wrist circumference).

3. Model Formulation

3.1 Linear Regression

Linear Regression estimates coefficients by minimizing the residual sum of squares between observed and predicted target values:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$
$$\min_{\beta} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

3.2 Ridge (L2) Regression

Ridge modifies the above by introducing a penalty on large coefficient magnitudes:

$$\min_{\beta} \sum_{i=1}^m (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^n \beta_j^2$$

The regularization parameter λ controls the extent of shrinkage. A small λ behaves like Linear Regression, while a large λ heavily penalizes large weights.

4. Evaluation and Metrics

Model evaluation was performed using the test set, based on two metrics:

- **R² (Coefficient of Determination):** measures the proportion of variance in BF % explained by the model.
- **RMSE (Root Mean Squared Error):** represents the average magnitude of prediction errors, expressed in the same units as the target variable.

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_i)^2}, RMSE = \sqrt{\frac{1}{m} \sum_i (y_i - \hat{y}_i)^2}$$

The **validation R² and RMSE** were used for parameter tuning, and the **test metrics** reflect final model generalization.

5. Results and Discussion

5.1 Quantitative Performance

Model	R ² (Test)	RMSE (Test)
Linear Regression	0.809	3.08
Ridge Regression ($\alpha = 0.1$)	0.839	2.86

Ridge Regression shows a moderate improvement in both R² and RMSE over Linear Regression, demonstrating that regularization slightly enhances generalization by reducing overfitting.

5.2 Comparative Predicted vs Actual Plot (Linear vs Ridge Regression)

The plot in the code shows **predicted Body Fat % (y-axis)** against the **actual measured Body Fat % (x-axis)** for both Linear and Ridge Regression. The blue curve represents Linear Regression predictions, while the red dashed curve represents Ridge Regression predictions. The black dotted line corresponds to the **ideal y = x line**, which indicates perfect predictions.

Interpretation:

- Both models follow the identity line closely, showing that they capture the overall linear trend effectively.
- Linear Regression slightly deviates at higher body fat percentage values (above ~35%), tending to **underpredict obese individuals**.
- Ridge Regression reduces this deviation, with its predictions staying closer to the ideal line across all ranges.
- This smoother fit demonstrates Ridge Regression’s **stabilizing effect** under correlated features and its slightly superior predictive accuracy.

Overall, Ridge Regression generalizes better while maintaining interpretability.

5.3 Residual Distribution Plot

The residual distribution plot compares the **error distributions** (actual – predicted values) of both models. Each model's residuals were plotted as smooth probability density curves.

Interpretation:

- Both residual distributions are approximately symmetric about zero, confirming the absence of systematic bias.
- Linear Regression exhibits a **wider residual spread**, indicating slightly higher prediction variance.
- Ridge Regression's residual curve is **narrower and more sharply peaked**, suggesting that most of its predictions lie closer to the actual values.
- This concentration around zero demonstrates Ridge's reduced variance and improved stability.

Thus, the residual analysis reinforces the numerical results — Ridge Regression, by penalizing excessive coefficient magnitudes, better controls prediction error spread and yields smoother generalization.

6. Conclusion

The study successfully developed and compared two regression models for predicting body fat percentage using anthropometric data.

Key findings:

- Both Linear and Ridge Regression models achieved high explanatory power, with $R^2 > 0.80$.
- Ridge Regression (L2 regularization) consistently achieved **lower RMSE** and **tighter residual distribution**, indicating superior generalization and robustness.
- Visual analyses through the Comparative and Residual Distribution Plots confirmed these results, highlighting Ridge's improved predictive stability and resistance to overfitting.

In summary, for datasets with multicollinearity among physical measurements, Ridge Regression provides a better trade-off between accuracy and stability than plain Linear Regression.

Future work could extend this study by evaluating nonlinear regression models (e.g., Random Forest Regressor or SVR) or incorporating feature selection to further improve interpretability and performance.

References

1. Fede Soriano, *Body Fat Prediction Dataset*, Kaggle, 2023.
2. Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR, 12(85):2825–2830, 2011.
3. Montgomery, Peck, and Vining, *Introduction to Linear Regression Analysis*, Wiley, 2021.